



DISTRICT LEGAL SERVICES AUTHORITY (DLSA), PORBANDAR

Under Gujarat State Legal Services Authority (GSLSA) & National Legal Services Authority (NALSA)

Official Citizen Legal Aid Portal & Administrative Cadre Management System

A Full-Stack E-Governance Platform for Constitutional Access to Justice under Article 39A and the Legal Services Authorities Act, 1987

Python 3.12 / Flask 3.1

SQLAlchemy 2.0 ORM

Microsoft SQL Server & SQLite Dual-Engine

100% Gujarati & English Parity

Project Title: DLSA Porbandar Citizen Legal Aid Portal
Domain: E-Governance, Legal Tech & Public Administration
Jurisdiction: Porbandar District Court Complex, Gujarat
Operational Talukas: Porbandar, Ranavav, Kutiyana

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1. Executive Summary & Project Abstract

Access to justice is an indispensable guarantee of a modern constitutional democracy. Under Article 39A of the Constitution of India and the Legal Services Authorities Act, 1987, the State is obligated to ensure that opportunities for securing justice are not denied to any citizen by reason of economic or social disability.

In Porbandar District, Gujarat, geographic isolation between coastal fishing hamlets, rural agricultural villages, and the district seat at Porbandar city has historically created friction for citizens attempting to access free legal counsel. Distressed citizens were forced to physically visit court premises, frequently falling prey to middlemen or facing severe wage loss.

The **District Legal Services Authority (DLSA) Porbandar Web Portal** was engineered to completely overcome these barriers. The portal digitizes legal aid intake, provides instant Section 12 statutory validation, generates trackable grievance tokens, authenticates 44 certified Para Legal Volunteers (PLVs) with printable digital identity cards, catalogs 31 frontline deployment clinics across police

stations, prisons, and hospitals, schedules Legal Literacy Camps and Lok Adalats with volunteer duty assignment, and provides complete bilingual parity in English and Gujarati.



2. Constitutional & Statutory Framework



Article 39A, Constitution of India

"The State shall secure that the operation of the legal system promotes justice, on a basis of equal opportunity, and shall, in particular, provide free legal aid, by suitable legislation or schemes or in any other way, to ensure that opportunities for securing justice are not denied to any citizen by reason of economic or other disabilities."

The statutory enforcement of Article 39A is carried out through the **Legal Services Authorities Act, 1987 (Act No. 39 of 1987)**. DLSA Porbandar operates under the direct superintendence of the Gujarat State Legal Services Authority (GSLSA) and the National Legal Services Authority (NALSA).

Statutory Section 12 Entitlements

Section 12 specifies the categories entitled to 100% free legal assistance, court fees exemption, and panel advocate representation:

Section 12 Clause	Entitled Citizen Category	System Verification Criteria
Section 12(a)	Scheduled Castes (SC) and Scheduled Tribes (ST)	Category declaration & caste documentation
Section 12(b)	Victims of human trafficking or <i>begar</i>	Special crisis relief intake
Section 12(c)	Women and Children	Universal eligibility (no income cap)
Section 12(d)	Persons with Disabilities (PwD)	Disability benchmark certification
Section 12(e)	Victims of mass disaster, ethnic violence, flood, earthquake	Emergency relief triage
Section 12(f)	Industrial Workmen	Labor disputes & industrial claims
Section 12(g)	Persons in custody (Undertrials & Convicts)	Custodial tracking at District Jails
Section 12(h)	General indigent citizens with annual income < ₹3,00,000/-	Income self-declaration verification

3. Problem Statement & Grassroots Gap Analysis

Prior to the deployment of this platform, legal services administration in Porbandar was constrained by manual, physical bottlenecks:

- **Geographical Friction:** Indigent citizens residing in coastal or border villages (e.g., Madhavpur, 60 km away) incurred substantial bus fares and lost daily earnings to submit paper applications at the court front office.
- **Opacity & Impersonation:** Citizens had no independent means to verify the identity of Para Legal Volunteers, exposing vulnerable litigants to unauthorized touts.
- **Information Asymmetry:** Schedules for Lok Adalats, criminal remand assistance at police stations, and jail clinic hours were inaccessible to the public.
- **Linguistic Exclusion:** Legal terminology and government circulars were predominantly in English, alienating rural Gujarati-speaking litigants.

4. Project Objectives & Comprehensive Scope

Citizen Empowerment Objectives

- Instant online application submission 24/7 without court visits.
- Automated Section 12 criteria checker with real-time feedback.
- Multi-stage case tracking via unique tracking token ('DLSA-2026-XXXX').
- 100% Gujarati and English bilingual parity.

Administrative & Cadre Objectives

- Digitize 44 certified PLVs with verifiable digital identity cards.
- Map 31 frontline centers (police stations, courts, clinics, villages).
- Role-based case scrutiny, counsel allocation, and simulated SMS.
- Schedule Legal Literacy Camps & Lok Adalats with duty assignments.



5. System Requirements & Specifications

5.1 Technology Stack Matrix

Component Layer	Technology Chosen	Version	Rationale & Justification
Backend Runtime	Python	3.12 / 3.14	High-level readability, robust cryptography, native ODBC support.
Web Framework	Flask (WSGI)	3.1.0	Lightweight microframework, zero bloat, rapid prototyping, modular routing.
ORM Layer	SQLAlchemy	2.0.38	Decoupled persistence, declarative mapping, native connection pooling.
Primary Database	Microsoft SQL Server	2019 / 2022	T-SQL stored procedures, enterprise security, ACID compliance.
Fallback Database	SQLite 3	3.45+	Zero-configuration, zero-downtime failover for local evaluation/demo.
Frontend Engine	HTML5 / CSS3 / ES6	Modern Standards	Semantic accessibility, responsive grid, native client-side filtering.
UI Framework	Bootstrap 5.3	5.3.3	Mobile-first responsive utilities, accessible modals, flexbox layouts.
Templating Engine	Jinja2	3.1.5	Dynamic template inheritance, XSS auto-escaping, custom bilingual filters.



6. System Architecture & High-Level Design

The platform follows the standard **Model-Template-View (MTV)** pattern. The application layer handles request routing, business validation, and database abstraction, while Jinja2 coordinates bilingual rendering for client web browsers.



Architectural Block Diagram

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+-----+
| CLIENT PRESENTATION LAYER |
| Citizens (Mobile/Desktop Web) DLSA Judicial Officers & Staff (Admin) |
+-----+
| HTTP/HTTPS (REST & Form Submissions)
v
+-----+
| APPLICATION CONTROLLER (Flask 3.1) |
| * URL Routing (app.py) * Role-Based Access Control (@admin_req)|
| * Bilingual Translation Engine (| t) * Simulated Notification Gateway (SMS) |
| * Case Review & Counsel Assignment * Camps & Lok Adalat Event Scheduler |
+-----+
| SQLAlchemy 2.0 Engine Mapping
v
+-----+
| PERSISTENCE ABSTRACTION LAYER |
| (Dynamic Dual-Database Controller) |
| +-----+ |
| | Primary Enterprise | Local Fallback Engine | |
| | Microsoft SQL Server | SQLite 3 (dlsa.db) | |
| | (pyodbc / T-SQL) | (Zero-Config Mode) | |
| +-----+ |
+-----+

```

Dual-Engine Database Failover Mechanism

The application implements a resilient database connection algorithm in **database.py**:

1. On startup, the system inspects the environment configuration (**DB_TYPE=auto**).
2. It attempts an ODBC handshake with Microsoft SQL Server using connection parameters from **.env**.
3. If SQL Server is reachable, it mounts the enterprise T-SQL database.
4. If the SQL Server service is offline, it intercepts the **pyodbc.OperationalError**, safely logs a diagnostic notice, and mounts local **dlsa.db** (SQLite).
5. Dynamic startup migration routine (**check_and_apply_migrations**) automatically executes non-destructive DDL modifications if newly added columns are detected.

7. Database Schema & Data Dictionary

The relational schema organizes judicial officers, certified PLVs, frontline clinics, citizen applications, schemes, and judicial schedules.

Entity Schemas & Relational Structure

Table Name	Primary Key	Foreign Keys	Core Business Purpose
members	id (INT)	None	Stores Chairman, Secretary, and Chief Legal Aid Defense Counsels.
plvs	id (INT)	None	Roster of 44 certified Para Legal Volunteers, badges, and contacts.
deployments	id (INT)	None	Catalog of 31 frontline clinics (police stations, courts, villages).
plv_assignments	id (INT)	plv_id, deployment_id	Many-to-many relationship linking volunteers to specific clinics.
legal_aid_applications	id (INT)	None	Citizen applications with Section 12 criteria, tracking token, status.
schemes	id (INT)	None	Statutory NALSA & GSLSA welfare schemes directory.
lok_adalat_events	id (INT)	None	Upcoming Lok Adalats & literacy camps with assigned PLV lists.
admin_users	id (INT)	None	Administrative staff accounts with PBKDF2-SHA256 salted hashes.

8. Core Subsystems & Technical Implementation

8.1 Citizen Legal Aid Application & Section 12 Verification (/apply)

The citizen application module ensures that distressed citizens can submit grievances under Section 12 without traveling to court premises. The server-side controller generates a unique tracking number, writes the record with status **Submitted**, and dispatches a

simulated SMS notification.

8.2 Real-Time Application Tracking Stepper (`/track`)

Litigants can enter their tracking token (e.g. **DLSA-2026-9135**) to inspect the 4-stage visual progress stepper:



1. Submitted → 2. Under Scrutiny → 3. Counsel Assigned → 4. Completed / Disposed

8.3 44 Certified PLVs & Digital Identity Card Generator (`/plv-portal`)

The system stores all 44 certified PLVs of Porbandar district. Clicking any profile generates a modal replicating the official DLSA Gujarat Identity Card with registration code, training accreditation, issue authority, and verified phone contact.

8.4 31 Frontline Deployment Locations & Clinics (`/deployments`)

Mappable directory categorized into 9 Police Stations (monitoring *D.K. Basu* arrest rights), 9 Court & Institutional Clinics (including Bhavsinhji General Hospital and District Jail), and 13 Village Legal Desks.

8.5 Camps & Lok Adalat Event Management Module (`/admin/camps`)

Administrative interface empowering staff to publish upcoming Lok Adalats and rural literacy camps. Administrators can specify the date, venue, target beneficiaries, legal topics, and dynamically multi-select assigned PLVs from the active roster.

9. UI/UX Design, Visual Language & Localization

- **Official Judicial Palette:** High-contrast Navy Blue (`#0f2b48`), Gold (`#c59b27`), and Slate Grey (`#eae2f2`) matching Indian Judiciary visual guidelines.
- **Unified Card System:** 20px border radius, dual-layer drop shadows (`0 4px 15px rgba(15,43,72,0.08)` and `3px 6px 12px rgba(15,43,72,0.06)`), eliminating disjointed interior white patches.
- **Native Bilingual Engine:** Server-side Jinja filter (`| t`) supported by a 77,000-byte verified Gujarati translation dictionary in `translations.py` with cookie persistence.
- **Scrollable Modals:** Strict Bootstrap 5 modal design placing forms directly as `modal-content` inside `modal-dialog-scrollable` with docked footers.

10. Security, Cryptography & Data Protection

- **Authentication:** PBKDF2 password hashing with SHA-256 via Werkzeug security library.
- **Authorization:** Role-based access control protecting all `/admin/*` routes via `@admin_required` decorator.
- **SQL Injection Immunity:** 100% parameterization via SQLAlchemy 2.0 ORM expressions.
- **Confidentiality:** Citizen dispute summaries shielded behind authenticated administrative views.

12. Project Outcomes, Community Impact & Future Roadmap

Direct Community Outcomes

- Eliminated travel costs and lost daily wages for rural Porbandar applicants.

- Provided 24/7 transparency into panel counsel assignment and status.
- Established official fraud prevention via certified PLV digital identity cards.
- Delivered complete linguistic inclusion in Gujarati.

Future Scalability Roadmap

- Meta Cloud WhatsApp Chatbot integration for voice & vernacular intake.
- Live bidirectional API link with central NALSA Portal (`nalsa.gov.in`).
- Biometric and QR-code geofenced attendance for field PLVs.
- AI-powered legal triage assistance using Google Gemini models.

13. Conclusion & Statutory References

The **District Legal Services Authority (DLSA) Porbandar Citizen Legal Aid Portal** illustrates how modern software engineering and open web standards can empower marginalized citizens and uphold the constitutional guarantee of Article 39A. By combining intuitive citizen access with structured administrative governance, the platform offers a replicable model for judicial digitalization across Gujarat and India.

Statutory References & Precedents

1. **The Constitution of India** (Articles 14, 21, and 39A).
2. **The Legal Services Authorities Act, 1987 (Act No. 39 of 1987)** (Sections 6, 9, 10, 11, 12, 19, 20).
3. **National Legal Services Authority (Free and Competent Legal Services) Regulations, 2010.**
4. **NALSA Scheme for Para-Legal Volunteers (Revised).**
5. *D.K. Basu v. State of West Bengal (1997) 1 SCC 416* (Statutory safeguards regarding arrest and legal representation).
6. *Hussainara Khatoon v. Home Secretary, State of Bihar (1979) 2 SCC 81.*
7. *Suk Das v. Union Territory of Arunachal Pradesh (1986) 2 SCC 401.*

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DLSA Porbandar Web Engineering Capstone Project